Project 4 - Gravity wave identification

The attached file contains mock time series strain data containing a signature that resembles gravitational wave and some noise on top of the signal. The analytical template for the signal is

$$h(t) = \alpha \exp(t)\{1 - \tanh[2(t - \beta)]\} \sin(\gamma t) \tag{1}$$

α, β, γ are parameters signifying some physical properties of the system involved. They are in the range

$$0 < \alpha < 2 \tag{2}$$

$$1 < \beta < 10 \tag{3}$$

$$1 < \gamma < 20 \tag{4}$$

Perform a random walk in this 3D parameter space $\theta \equiv (\alpha, \beta, \gamma)$ using the Metropolis-Hastings algorithm. The probability of a particular choice of parameters is proportional to

$$P(\theta|data) \propto P(data|\theta)P(\theta) \tag{5}$$

The prior $P(\theta)$ is uniform in the range given above. The $P(data|\theta)$ is estimated as

$$P(data|theta) \propto \exp(\mathcal{Y}) \tag{6}$$

$$\mathcal{Y} = -\sum_i \frac{(ydata_i - ymodel_i)^2}{yerr_i^2} \tag{7}$$

where $ydata_i$ is the model data at time point t_i and $ymodel_i$ is the model prediction. Assume an error of 20% at each data point. Plot the probability distributions of each parameter obtained from the Metropolis algorithm.